



College of Computing and Informatics
Department of Computer Science

Course Syllabus

Course Title

Design & Analysis of Algorithms

Course Code
1501371

Fall Semester 2024-2025

Course Syllabus

Course Title	Design & Analysis of Algorithms		
Course Code	1501371		
Credit Hours	3		
Pre-requisite(s)	1501279 Discrete Structures, Data structure 1501216		
Co-requisite(s)	None		
Semester	Fall	Year	24/25
Instructors Name	Dr. Ahmed Khedr		
Office Location	W5-226		
Tel. No.	3560		
Email	akhedr@sharjah.ac.ae		
Lecture Times	TR 8.00-9.15		
Office Hours	MW 1.00-3.00pm		

Course Description (as in the catalogue):

This course emphasizes on the fundamental concepts of analyzing and designing algorithms, including divide and conquer, greedy methods, backtracking, randomization and dynamic programming. A number of algorithms for solving problems which arise often in applications of computer science are covered, including sorting, searching, graph algorithms, maximum flow, and string matching. Also, as a part of this course, the NP, NP-complete, and NP-hard Problems and an introduction to Parallel and distributed algorithms will be covered.

Course Objectives/Goals:

The goals of this course are to enable students to:

- Provide theoretical background and thinking ability for analyzing and designing algorithms.
- Introduce different algorithm design paradigms: divide and conquer, greedy methods, randomization and dynamic programming.
- Introduce how to analyze the correctness and find runtime complexity (upper and lower bounds) of a given algorithm.
- Use of these concepts and techniques to solve practical problems.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

1. Evaluate the efficiency of an algorithm.
2. Determine and apply the appropriate design Technique (i.e., Divide & Conquer, Greedy, Dynamic Programming, and Backtracking algorithms) to solve a given problem.
3. Employ graphs & trees representations to model and solve problems in the field.
4. Analyze strings and pattern matching algorithms, their time complexity and applications.
5. Compare P and NP complexities.

Alignment of Course Student Learning Outcomes to the BSc-CS Program Student Learning Outcomes

Program SLOs	Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	- Evaluate the efficiency of an algorithm. - Analyze strings and pattern matching algorithms, their time complexity and applications. - Compare P and NP complexities.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	- Evaluate the efficiency of an algorithm. - Employ graphs & trees representations to model and solve problems in the field.
Communicate effectively in a variety of professional contexts.	-
Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.	-
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.	-
Apply computer science theory and software development fundamentals to produce computing-based solutions.	- Evaluate the efficiency of an algorithm. - Determine and apply the appropriate design Technique (i.e., Divide & Conquer, Greedy, Dynamic Programming, and Backtracking algorithms) to solve a given problem. - Employ graphs & trees representations to model and solve problems in the field. -

Alignment of Course Student Learning Outcomes to the BSc-BI Program Student Learning Outcomes

Program SLOs	Course SLOs
Apply knowledge of mathematics, computer science, computer engineering, and computational biology appropriate to the discipline.	- Determine and apply the appropriate design Technique (i.e., Divide & Conquer, Greedy, Dynamic Programming, and Backtracking algorithms) to solve a given problem. - Employ graphs & trees representations to model and solve problems in the field.
Analyze a problem, identify and define the computing requirements appropriate to its solution.	- Evaluate the efficiency of an algorithm. - Analyze strings and pattern matching algorithms, their time complexity and applications. - Compare P and NP complexities.
Design and evaluate a software system, process, or program to meet desired needs.	- Evaluate the efficiency of an algorithm. - Employ graphs & trees representations to model and solve problems in the field.
Function effectively on teams and communicate effectively with a range of audiences.	-
Identify professional, ethical, legal, security and social issues and responsibilities related to biomedical informatics.	-
Analyse the local and global impact of biomedical informatics on individuals, organizations, and society.	-
Apply design and development principles in the construction of software and informatics- based solutions of varying complexity.	- Determine and apply the appropriate design Technique (i.e., Divide & Conquer, Greedy, Dynamic Programming, and Backtracking algorithms) to solve a given problem.

Weekly Distribution of Course Topics/Contents

Week	Topic	Comment*	Course SLO
1.	Introduction	Assignment1	1,2
2.	Growth Functions (Big and small oh, Big and small Theta, Big and small Omega)		
3.	Recurrence relations (Substitution, Iterative and Master methods to solve the recurrence relations).		
4.	Incremental strategy and Searching.		
5.	Divided and Conquer, Sorting (Selection Sort, merge sort, quick sort, Heap sort, etc.), Sorting in Linear Time such as counting sort and Radix sort.	Assignment2	1,2
6.	Order statistics (i^{th} smallest element).		
7.	Midterm review and Exam.	Assignment3	1,2,3,4
8.			

Week	Topic	Comment*	Course SLO
9.	Greedy Algorithms, Graph Algorithms (Searching graphs, Minimum Spanning Trees, Shortest/Longest Paths). Applications of Minimum Spanning Trees, Shortest/Longest Paths.		
10.	Dynamic Programming (DP), String matching, Backtracking (BT), randomization algorithms, differences between DP and BT and their applications.		
11.			
12.			
13.	Maximum Flow (Ford Fulkerson algorithm, Karp Algorithm and minimum cut maximum flow theory). Applications of Maximum flow (Problems can be solved using maximum flow).	Assignment4	1,3,5
14.	Parallel and distributed Algorithms (PRAM algorithm Distributed algorithms).		
15.	P, NP, NP-complete, NP- Hard problems.		
16.	Final Exam		

* In the comments, you can add the relevant chapter or notes, etc.

Information on out-of-class assignments with due dates for submission

Assignment/Activity	Due Date	Comments
Assignment1	Week 3	
Assignment3	Week 6	
Assignment3	Week 9	
Assignment4	Week 12	

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Mid-Term	Week 8	25
Quizzes	Weeks 3, 9 and 12	15
Final Exam	Week 16	45
Assignments	See table above	15
Total		100

** You can modify / add other tools relevant to the course.

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
Evaluate the efficiency of an algorithm.	- Lecture - Problem solving	- Assignments - Quizzes	Marks

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
Determine and apply the appropriate design Technique (i.e., Divide & Conquer, Greedy, Dynamic Programming, and Backtracking algorithms) to solve a given problem.	- Lecture - Problem solving	- Assignments - Quizzes - Exams	Marks
Employ graphs & trees representations to model and solve problems in the field.	- Lecture - Problem solving	- Assignments - Quizzes - Exams	Marks
Analyze strings and pattern matching algorithms, their time complexity and applications.	- Lecture - Problem solving	- Assignments - Quizzes	Marks
Compare P and NP complexities.	- Lecture - Problem solving	- Assignments - Quizzes - Exams	Marks

Teaching and Learning Resources:

Text Book(s):

Dasgupta, S., Papadimitriou, C. and Vazirani, U. (2011). Algorithms. McGraw-Hill Education - Europe.

Other Resources:

- Cormen, et al. (2009). Introduction to Algorithms. 3rd Edition. The MIT PRESS.
- Neapolitan, R. (2014). Foundations of Algorithms. 5th Edition. Jones & Bartlett Learning

Attendance policy:

Lectures attendance are required. Students are expected to come to class on time and stay for the whole period. Warnings will be given according to the University policy. A student missing 10% of the total allocated course hours will be sent the 1st warning notice and a student missing 15% will be sent the 2nd warning notice. A student missing 20% will be barred from attending the final exam.

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

In-class Activities

Students are expected to do their own work individually and/or in teams. Your work individually and in teams will be graded. Activities include quizzes, individual assignments, and team assignments.

Teaching and Learning:

The course includes weekly lectures, quizzes, and a number of assignments.

Academic Integrity, Collaboration

All students are expected to maintain the utmost level of academic integrity. Any attempts to cheat will lead to appropriate sanctions. Home works should be solved mostly individually. Substantial collaboration, in particular in writing up solutions or during programming assignment, is not appropriate. Nor is of course copying classmates' solutions. It is also never appropriate to search for solutions online and try to pass them off as your own work.

You may discuss problems with your fellow students and the course staff, but you must write your own solutions. You may also consult any publications. Clearly cite any collaborators or outside sources. You may not give or receive help from anyone or any publications.